

ASSIGNMENT NO.2: Basic Data Retrieval using SQL

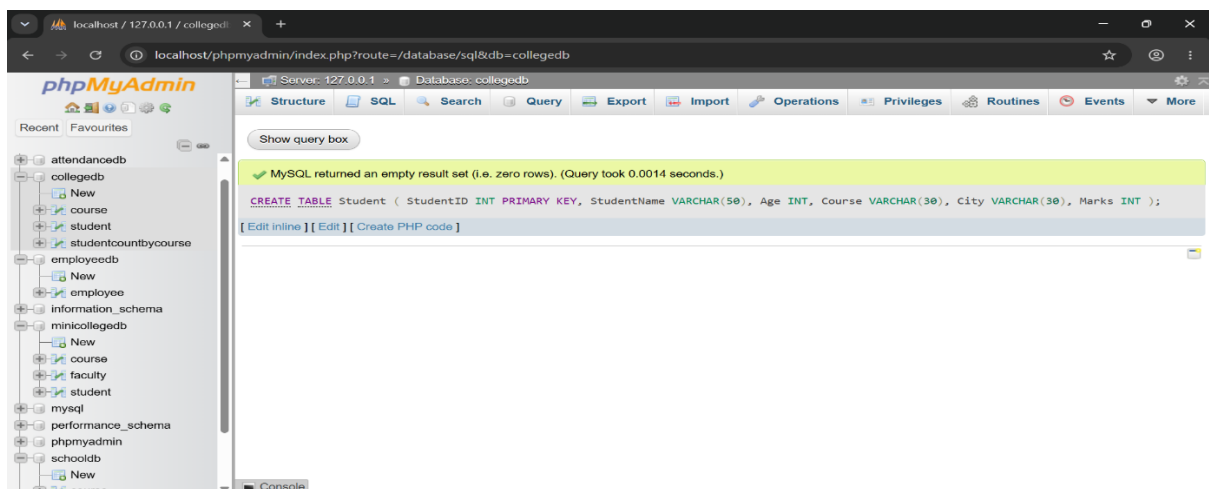
Questions:

Q1) Create a database named CollegeDB and a table named Student. The table should contain the following columns:

Column Name	Data Type
StudentID	INT
StudentName	VARCHAR(50)
Age	INT
Course	VARCHAR(30)
City	VARCHAR(30)
Marks	INT

Add a primary key on StudentID and insert at least 10 records.

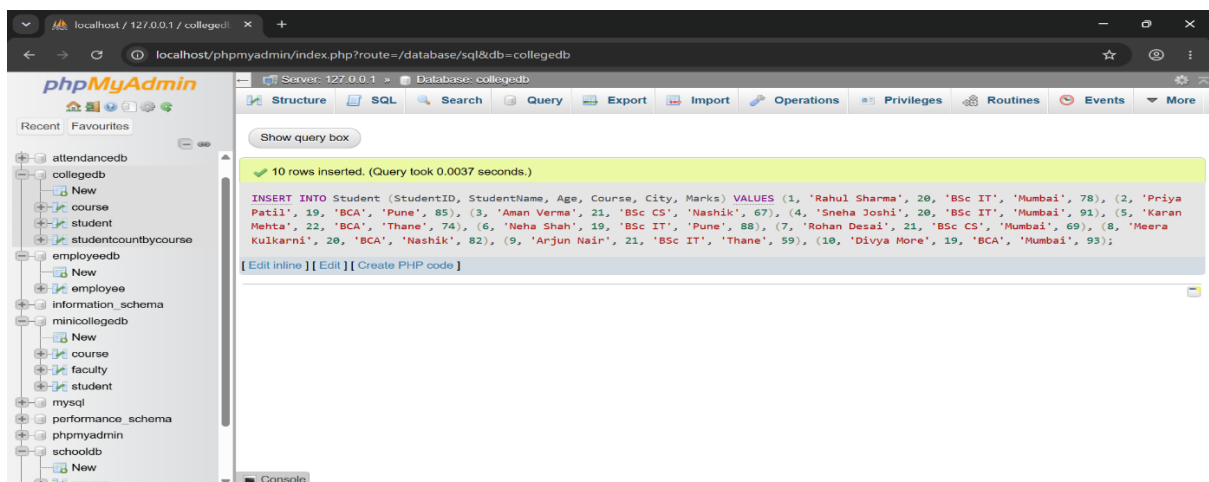
Answer:-



The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The SQL tab is active, displaying the following query:

```
CREATE TABLE Student ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Age INT, Course VARCHAR(30), City VARCHAR(30), Marks INT );
```

The execution result shows: "MySQL returned an empty result set (i.e. zero rows). (Query took 0.0014 seconds.)"



The screenshot shows the phpMyAdmin interface with the 'collegedb' database selected. The SQL tab is active, displaying the following query:

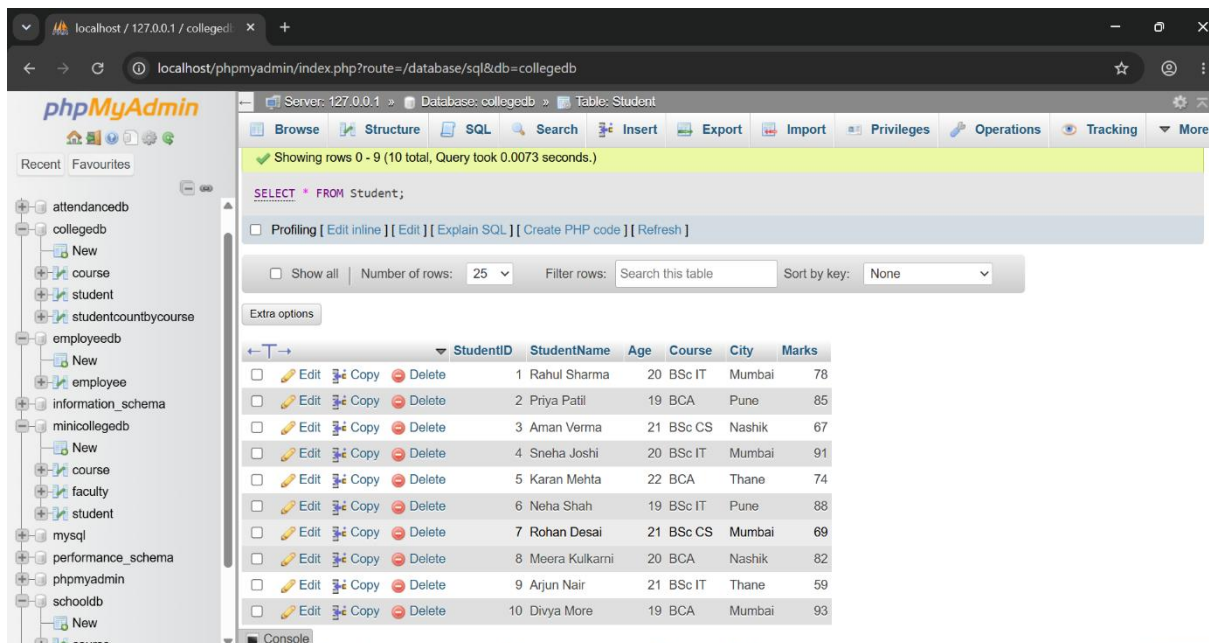
```
INSERT INTO Student (StudentID, StudentName, Age, Course, City, Marks) VALUES (1, 'Rahul Sharma', 20, 'BSc IT', 'Mumbai', 78), (2, 'Priya Patil', 19, 'BCA', 'Pune', 85), (3, 'Aman Verma', 21, 'BSc CS', 'Nashik', 67), (4, 'Sneha Joshi', 20, 'BSc IT', 'Mumbai', 91), (5, 'Karan Mehta', 22, 'BCA', 'Thane', 74), (6, 'Neha Shah', 19, 'BSc IT', 'Pune', 88), (7, 'Rohan Desai', 21, 'BSc CS', 'Mumbai', 69), (8, 'Meera Kulkarni', 20, 'BCA', 'Nashik', 82), (9, 'Arjun Nair', 21, 'BSc IT', 'Thane', 59), (10, 'Divya More', 19, 'BCA', 'Mumbai', 93);
```

The execution result shows: "10 rows inserted. (Query took 0.0037 seconds.)"

Q2) Write a MySQL query to display all records from the Student table.

The output should display all columns and all rows.

Answer:-

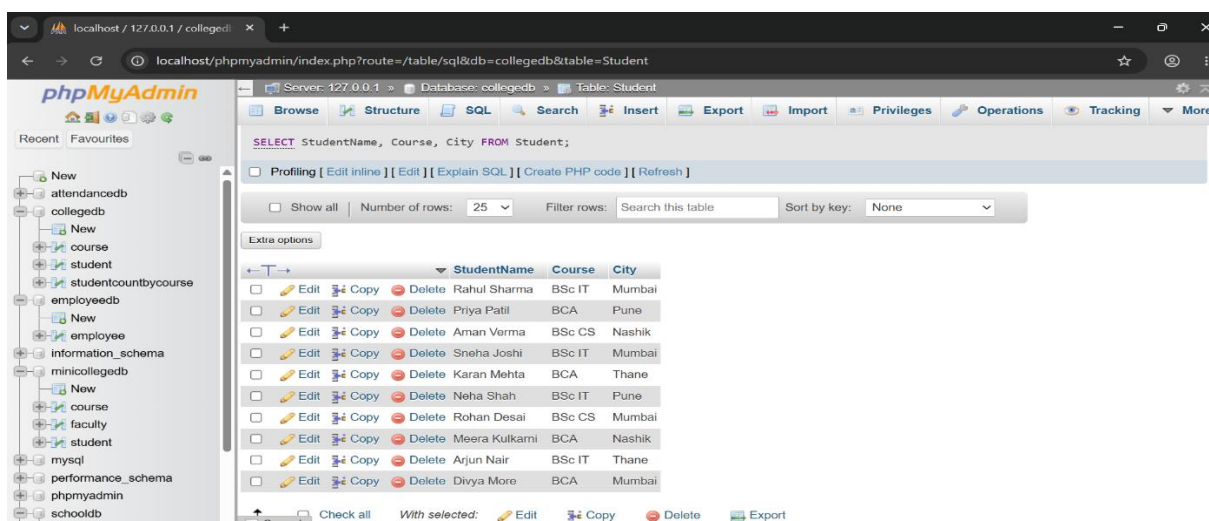


The screenshot shows the phpMyAdmin interface for a database named 'collegedb'. The 'Student' table is selected, and the SQL query 'SELECT * FROM Student;' is entered. The query has been executed, displaying 10 rows. The table structure is as follows:

StudentID	StudentName	Age	Course	City	Marks
1	Rahul Sharma	20	BSc IT	Mumbai	78
2	Priya Patil	19	BCA	Pune	85
3	Aman Verma	21	BSc CS	Nashik	67
4	Sneha Joshi	20	BSc IT	Mumbai	91
5	Karan Mehta	22	BCA	Thane	74
6	Neha Shah	19	BSc IT	Pune	88
7	Rohan Desai	21	BSc CS	Mumbai	69
8	Meera Kulkarni	20	BCA	Nashik	82
9	Arjun Nair	21	BSc IT	Thane	59
10	Divya More	19	BCA	Mumbai	93

Q3) Write a MySQL query to display only the following columns:

1. StudentName
2. Course
3. City



The screenshot shows the phpMyAdmin interface for the 'Student' table. The SQL query 'SELECT StudentName, Course, City FROM Student;' is entered. The query has been executed, displaying 10 rows with only the specified columns. The table structure is as follows:

StudentName	Course	City
Rahul Sharma	BSc IT	Mumbai
Priya Patil	BCA	Pune
Aman Verma	BSc CS	Nashik
Sneha Joshi	BSc IT	Mumbai
Karan Mehta	BCA	Thane
Neha Shah	BSc IT	Pune
Rohan Desai	BSc CS	Mumbai
Meera Kulkarni	BCA	Nashik
Arjun Nair	BSc IT	Thane
Divya More	BCA	Mumbai

Q4) Write a MySQL query to display students who belong to the city Mumbai.

The output should include:

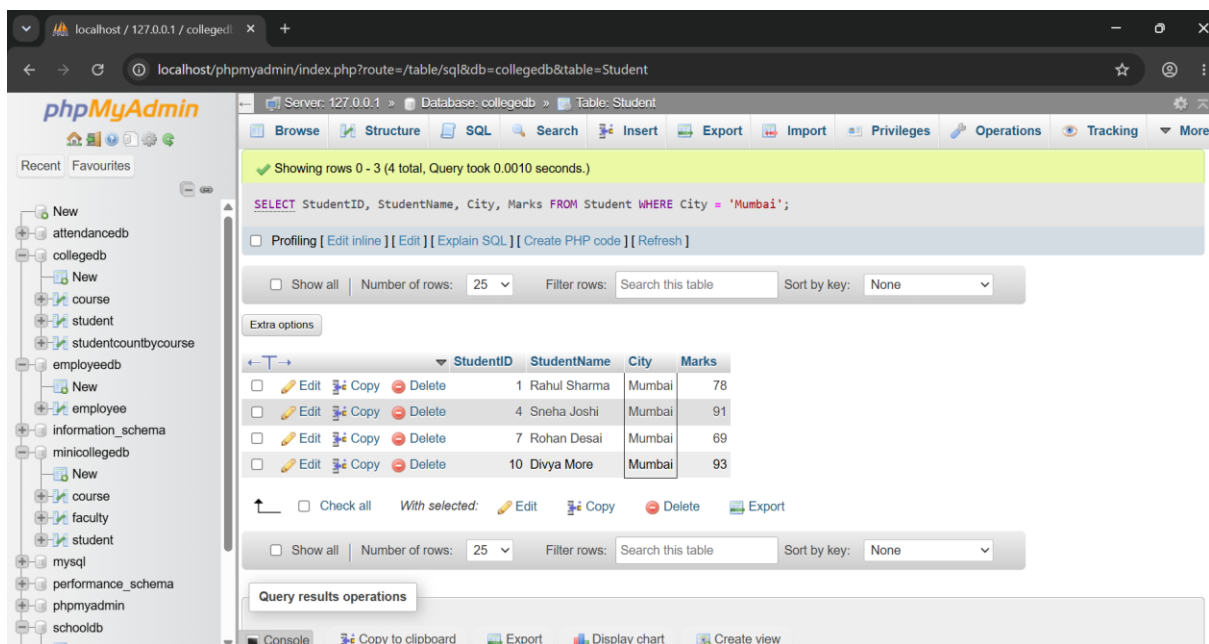
1. StudentID
2. StudentName
3. City
4. Marks

MYSQL query:-

SELECT StudentID, StudentName, City, Marks

FROM Student

WHERE City = 'Mumbai';



Showing rows 0 - 3 (4 total, Query took 0.0010 seconds.)

```
SELECT StudentID, StudentName, City, Marks FROM Student WHERE City = 'Mumbai';
```

Number of rows: 25 Filter rows: Search this table Sort by key: None

	StudentID	StudentName	City	Marks
☐	1	Rahul Sharma	Mumbai	78
☐	4	Sneha Joshi	Mumbai	91
☐	7	Rohan Desai	Mumbai	69
☐	10	Divya More	Mumbai	93

Query results operations

Q5) Write a MySQL query to display students whose marks are greater than 80.

The output should be sorted by marks in descending order.

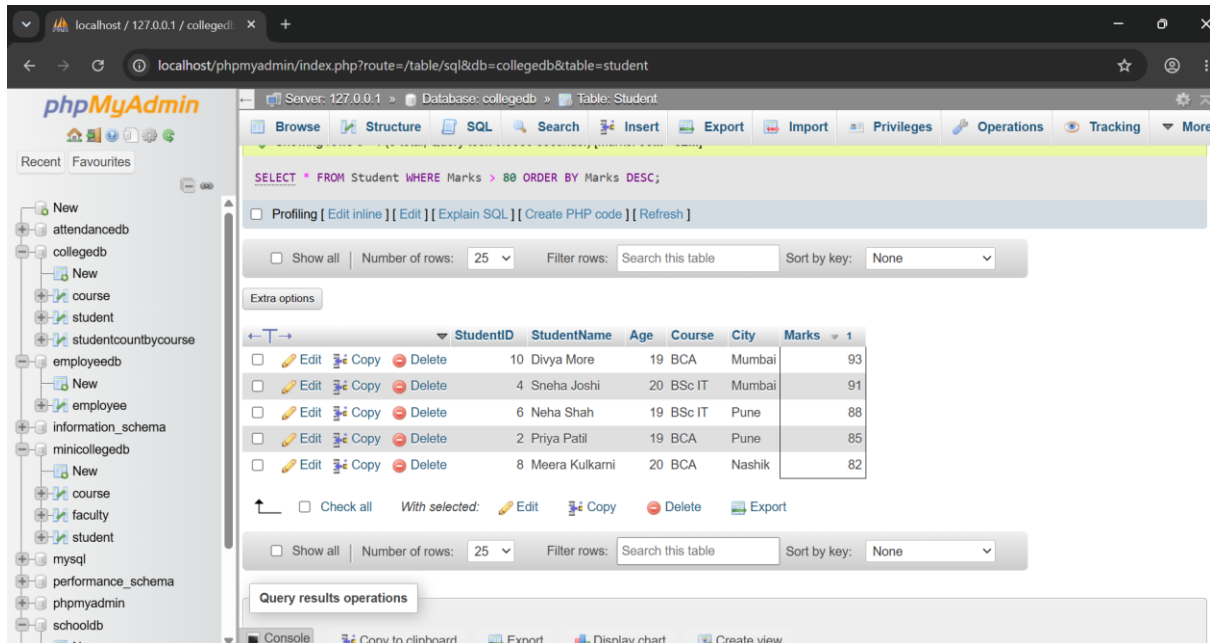
MySQL:-

SELECT *

FROM Student

WHERE Marks > 80

ORDER BY Marks DESC;



Q6) Write a MySQL query to display the top 3 students based on marks.

The query should:

1. Display StudentName and Marks.
2. Sort marks from highest to lowest.
3. Display only 3 records using LIMIT.

MySQL:-

SELECT StudentName, Marks

FROM Student

ORDER BY Marks DESC

LIMIT 3;

localhost / 127.0.0.1 / collegedi x +

localhost/phpmyadmin/index.php?route=/table/sql&db=collegedb&table=Student

phpMyAdmin

Recent Favourites

Server: 127.0.0.1 Database: collegedb Table: Student

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Showing rows 0 - 2 (3 total, Query took 0.0015 seconds.) [Marks: 93... - 88...]

`SELECT StudentName, Marks FROM Student ORDER BY Marks DESC LIMIT 3;`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Extra options

	StudentName	Marks
☐ Edit ☐ Copy ☐ Delete	Divya More	93
☐ Edit ☐ Copy ☐ Delete	Sneha Joshi	91
☐ Edit ☐ Copy ☐ Delete	Neha Shah	88

☐ Check all With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

Query results operations

☐ Print ☐ Copy to clipboard ☐ Export ☐ Display chart ☐ Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Console